

Sai Srikar Chirumarri

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Education

- 2024 - 2025 **MSc. Computer Games Technology with VR**
City University of London, London, England.
- 2022 - 2024 **B.tech CSE with specialization in Robotics**
Vellore institute of technology, India.
CGPA: 8.4

Skills and Certifications

- **Programming:** C++/C, C, Java, Python
- **Frameworks/Tools:** Visual Studio, Git, Unreal Engine, Unity.
- **Tools:** Blender, Quixel mixer, Substance Painter, Maya, Marvelous Designer
- **Certifications:** Unreal Engine 5 C++ Developer, Data Structures and OOP with C++

Projects

1) The Buckerberg Files (Unity, C, VR)

- **Role:** Gameplay Programmer Level Designer (Team of 5).
- **Description:** An immersive VR puzzle game exploring data privacy. Developed core VR systems including hand presence, physical interactions (pockets, breakables, etc), and haptic feedback.
- **Key Contributions:**
 - i) Implemented full VR player setup and environmental interaction system.
 - ii) Designed branching narrative paths and VR-specific UI (menus, end-screen).
 - iii) Integrated spatial audio and haptics to enhance immersion.

2) Shackled Shadow (Monogame, C, 2D)

- **Role:** Sole Developer.
- **Description:** A 2D puzzle-platformer where the player, a sentient shadow, switches between worlds to solve puzzles and fight enemies.
- **Key Contributions:**
 - i) Built the game from scratch, implementing core systems: game state management, input handling, and score tracking.
 - ii) Created an innovative core mechanic: dynamic world-switching that alters level art and gameplay.
 - iii) Designed a data-driven level system parsed from TXT files for efficient iteration.

3) Dynamic NPC System (Unreal Engine, C++, VR)

- **Role:** Sole Developer (Masters Dissertation Project).
- **Description:** A research project to enhance VR immersion through NPCs that react realistically to physical touch.

- **Key Contributions:**

- i) Architected a custom collision system to detect touch location and intensity on NPC body parts.
- ii) Designed an AI Behavior Tree and FSM that drives context-aware reactions (Aggressive, Friendly).
- iii) Implemented a physics-based animation reaction system that blends procedural bone movement with ragdoll physics and also features bone grab system, built as a reusable Unreal plugin.

4) Planet Clearance (Custom Game Engine, C++, 3D)

- **Role:** Sole Developer.

- **Description:** A third-person shooter featuring a custom game engine built with C++ and OpenGL.

- **Key Contributions:**

- i) Engineered a custom 3D game engine, rendering primitive objects and UI with OpenGL.
- ii) Implemented core gameplay systems: third-person camera, NPC FSM (4 states), audio via FMOD, and visual effects (fog, explosions).
- iii) Developed power-ups and a dynamic crossfade system.